

v1.0Muon Deep Dive_ One-Particle Universe

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Muon Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: v1.0Muon Deep Dive_ One-Particle Universe V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Muon g-2 status	Preliminary FNAL result in V1.0	FNAL 2023 final: 5.1σ (KNT18), 1.5σ (BMW). V2.0 stores both; muon precession game mechanic reflects tension	HIGH	Definitive result + BMW controversy — V1.0 claims premature confirmation
Mu2e conversion	Not covered	Mu2e@Fermilab 2023: sensitivity to μ↔e with R < 10^-17. LFNV_AMPLITUDE tracking per trajectory	MEDIUM	New lepton flavor violation search mode added to simulation
Muon lifetime precision	τ = 2.197 μs	MuLan: τ = 2196980.3±2.2 ps □ G_F = 1.1663788×10^-5 GeV^-2. V2.0 weak coupling updated	LOW	Sub-ppm precision reduces weak decay uncertainty
Muon catalyzed fusion	Not covered	μCF: muon replaces electron, 0.0053 Å bond length. Fusion rate 10^8 /s. V2.0 MUON_CATALYSIS mode added	LOW	Interesting simulation mode demonstrating muon mass effect

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Muon
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric ($C \cdot F \cdot S$ product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0 Muon Deep Dive_ One-Particle Universe V1.0 archived | V2.0 is the active specification as of June 2026